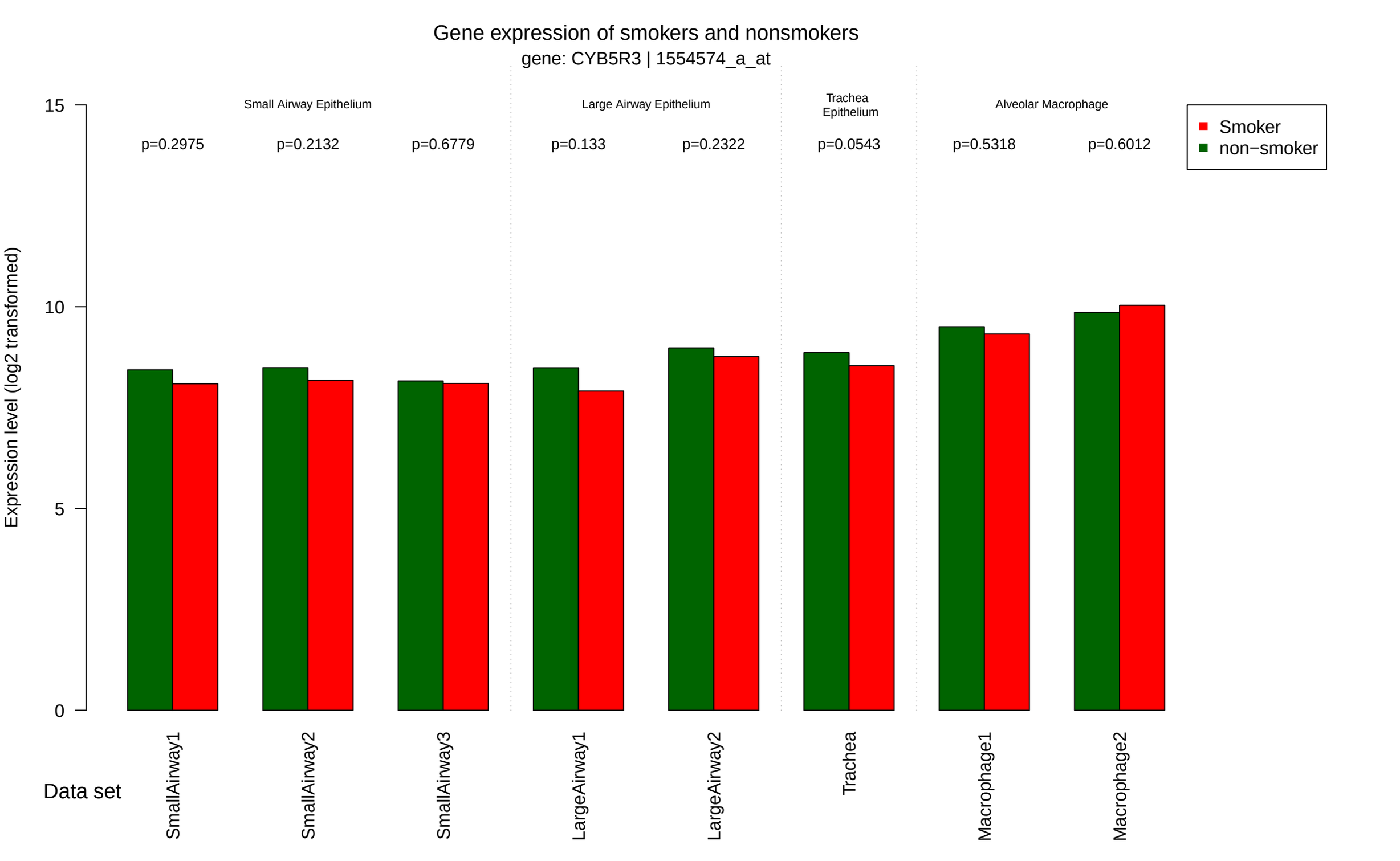
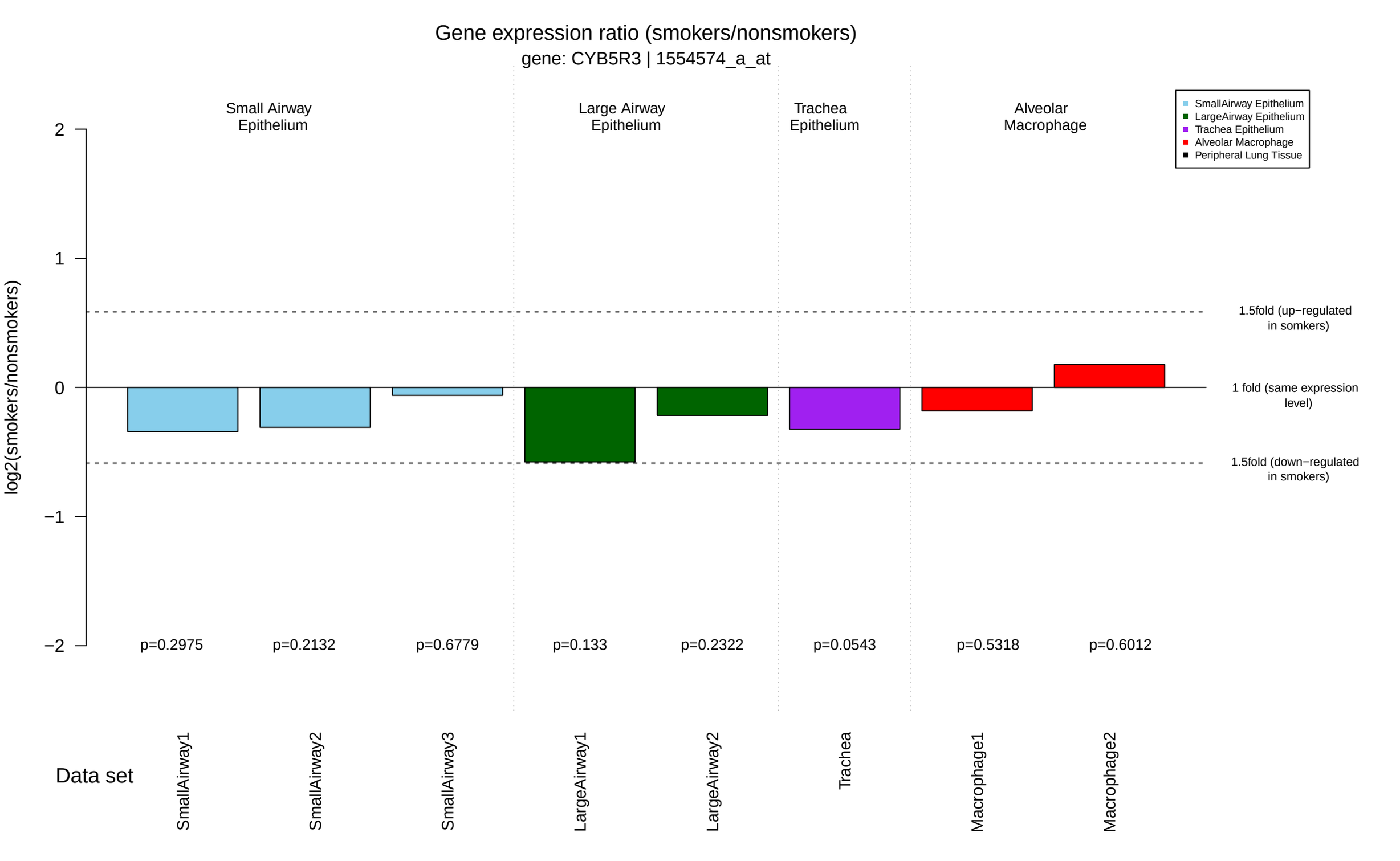
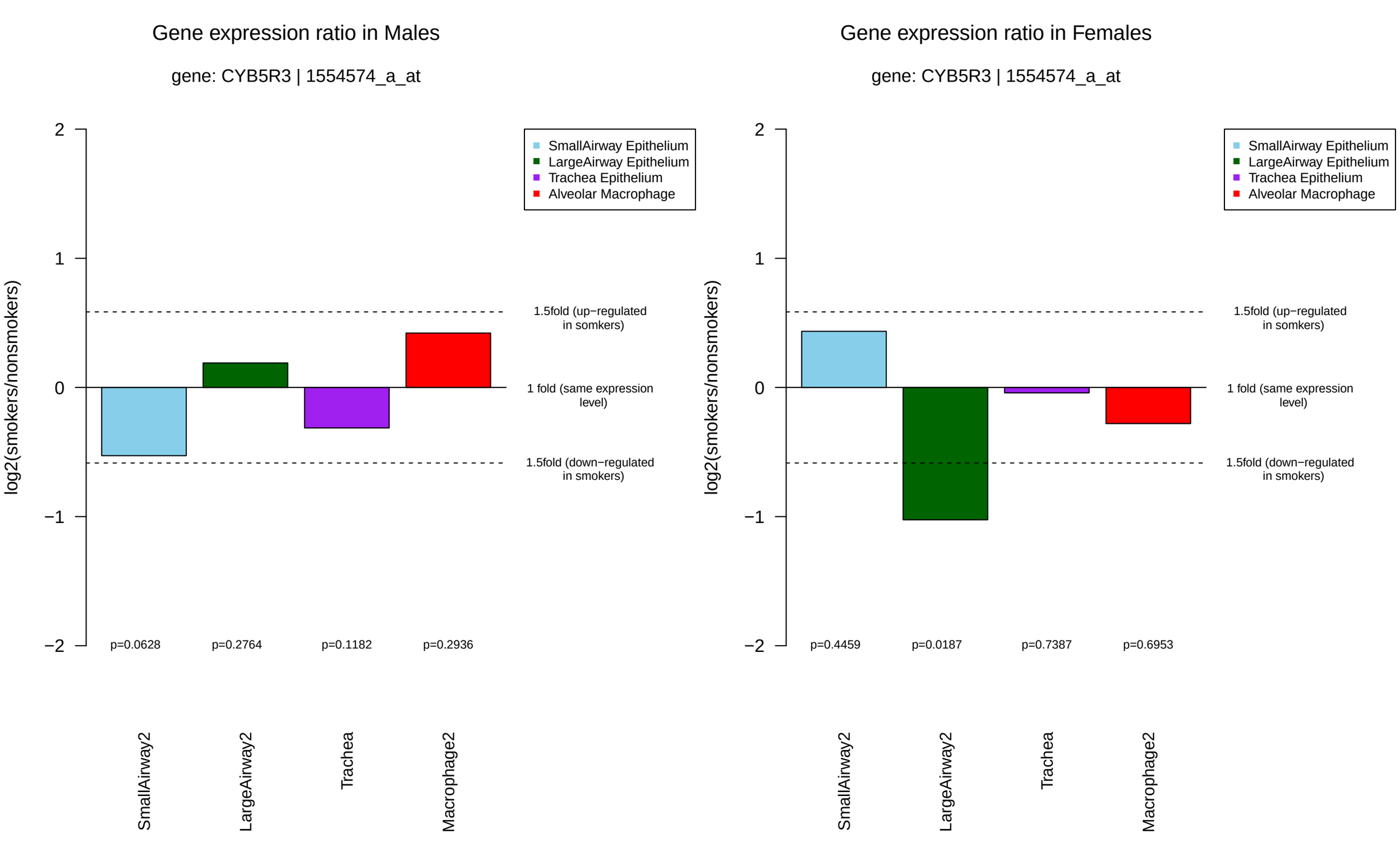
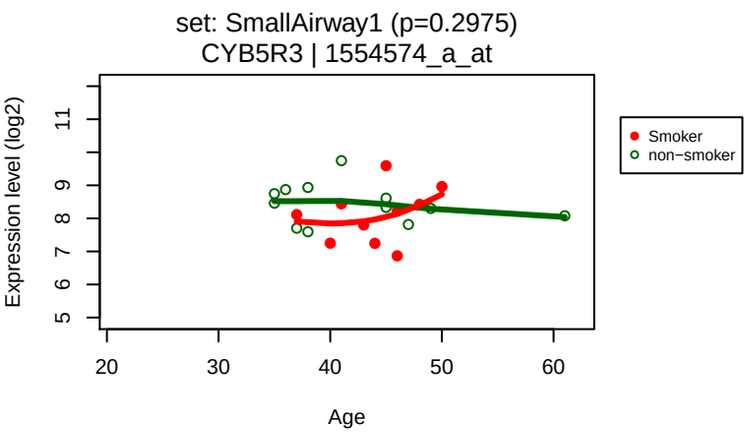
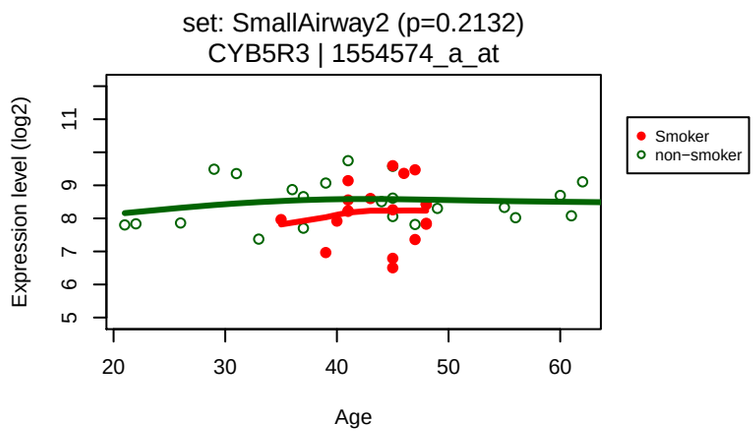
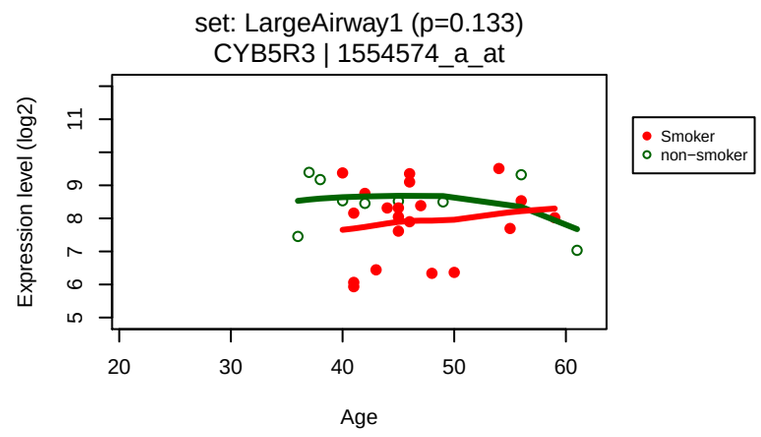

Gene expression of smokers and nonsmokers
gene: CYB5R3 | 1554574_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2975
p=0.2132
p=0.6779
p=0.133
p=0.2322
p=0.0543
p=0.5318
p=0.6012
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

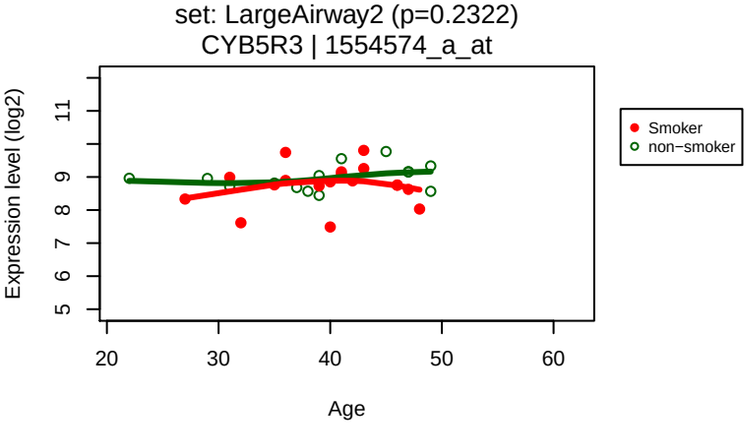

Gene expression ratio (smokers/nonsmokers)
gene: CYB5R3 | 1554574_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2975
p=0.2132
p=0.6779
p=0.133
p=0.2322
p=0.0543
p=0.5318
p=0.6012
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

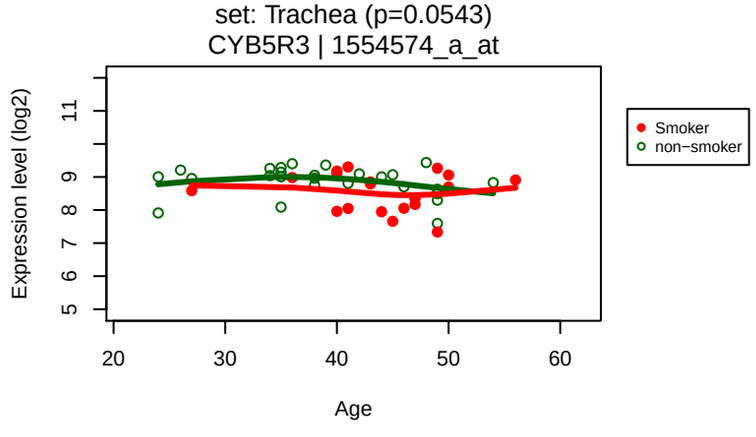

Gene expression ratio in Males
gene: CYB5R3 | 1554574_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0628
p=0.2764
p=0.1182
p=0.2936
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYB5R3 | 1554574_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4459
p=0.0187
p=0.7387
p=0.6953
SmallAirway2
LargeAirway2
Trachea
Macrophage2

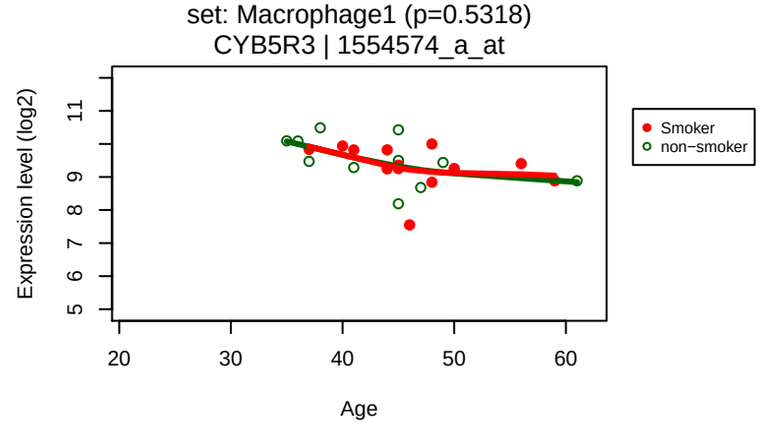

set: SmallAirway1 (p=0.2975)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker

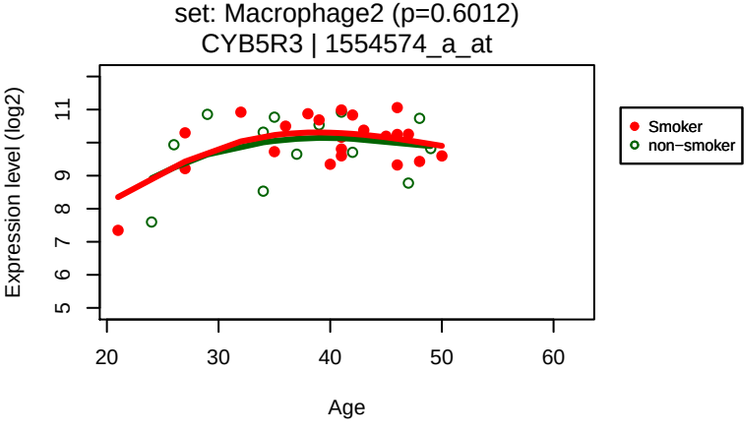

set: SmallAirway2 (p=0.2132)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker

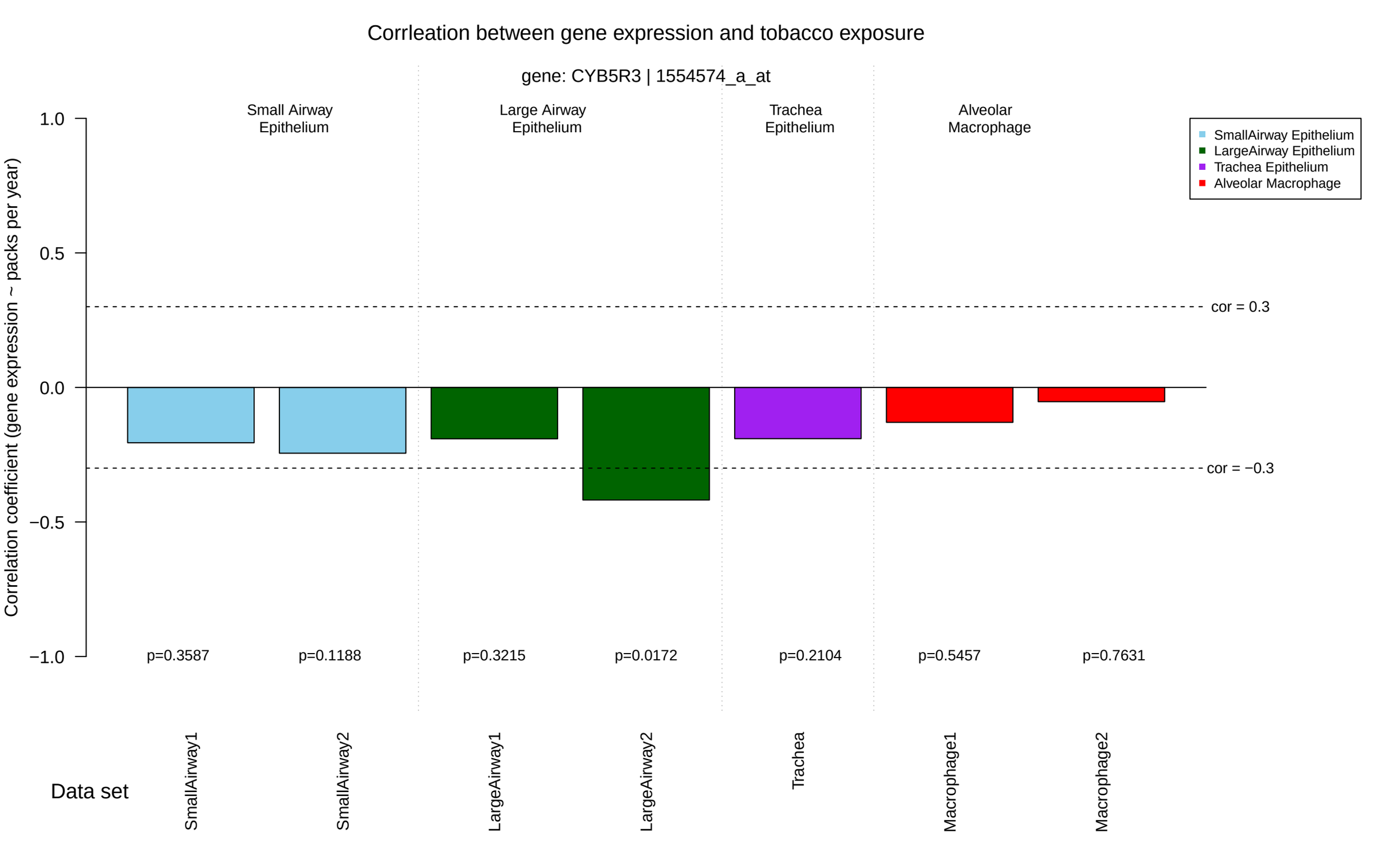

set: LargeAirway1 (p=0.133)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2322)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0543)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5318)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6012)
CYB5R3 | 1554574_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYB5R3 | 1554574_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3587
p=0.1188
p=0.3215
p=0.0172
p=0.2104
p=0.5457
p=0.7631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2